

NATTC Full Stack Development - Complete Notes (Roman Urdu)

1. Frontend (HTML, CSS, JavaScript, React)

Frontend wo hissa hota hai jo user dekhta aur use karta hai. Misal: Daraz ya Facebook ka logo, buttons, images, menu aur text sab frontend hain.

HTML website ka structure banata hai. Ghar ki deewaron aur frame ki tarah.

Example: `<h1>Welcome</h1> <p>Hello Nabeel</p>`

CSS website ko khoobsurat banata hai. Ghar ki paint aur decoration ki tarah.

JavaScript website ko interactive banata hai. Button click hone par action lena, form submit karna ya cart update karna JavaScript ka kaam hai.

React JavaScript ki library hai jo bari aur fast websites banane ke liye use hoti hai.

2. Backend (Node.js + Express)

Backend website ka asli dimagh hota hai. ATM ki screen frontend hai, lekin balance check aur paise nikalna backend karta hai.

Node.js JavaScript ko server par chalata hai. Express Node.js ka framework hai jisse Login, Signup aur Payment APIs banayi jati hain.

3. Database (SQL aur MongoDB)

Database data save karta hai. School records, bank accounts aur Facebook users ka data database mein hota hai.

SQL table format use karta hai. MongoDB JSON documents use karta hai.

4. Website

Website web pages ka collection hoti hai. Home, Profile, Settings aur Contact mil kar website banate hain.

5. Browser, Search Engine aur Internet

Browser: Chrome, Firefox.

Search Engine: Google, Bing.

Browser = Car, Search Engine = Driver, Internet = Road.

6. Network Types

LAN: Chhota network, jaise school ya office.

MAN: Pore city ka network.

WAN: Duniya bhar ka network, yani Internet.

7. Domain Name

Website ka naam, jaise google.com.

8. Hosting Server

Server ek powerful computer hota hai jahan website online store hoti hai.

9. DNS aur IP Address

DNS domain name ko IP address mein convert karta hai.

10. Client → Request → Server → Response

Restaurant example: User = Client, Order = Request, Kitchen = Server, Food = Response.

11. SDLC

Requirement → Planning → Design → Development → Testing → Deployment → Maintenance.

12. Requirement

Customer se poochna ke software mein kya features chahiye.

13. Planning

Language, team aur time decide karna.

14. Design

Website ka layout aur UI banana.

15. Development

Coding karna.

16. Testing

Errors aur bugs check karna.

17. Deployment

Website ko internet par live karna.

18. Maintenance

Updates aur bug fixes karna.

19. Web Application

Facebook, YouTube aur Daraz jahan user interact karta hai.

20. Static vs Dynamic Website

Static: Content fix hota hai.

Dynamic: Content user ke hisaab se badalta rehta hai.

21. MERN vs MEAN

MERN = MongoDB + Express + React + Node.js.

MEAN = MongoDB + Express + Angular + Node.js.

22. LMS (Learning Management System)

Google Classroom, Moodle, Udemy aur Coursera LMS ki examples hain. Teacher lectures upload karta hai aur student assignments submit karta hai.

23. HTML Element

Example: `<p>Hello Nabeel</p>`

Opening Tag + Content + Closing Tag mil kar HTML Element banta hai.

Real Life Example:

Salon Booking Website mein HTML structure banata hai, CSS design karta hai, JavaScript interaction deta hai, React frontend manage karta hai, Node.js aur Express backend chalte hain, MongoDB data save karta hai, Server website ko online rakhta hai, Domain naam deta hai aur DNS us naam ko IP mein convert karta hai.